



Question Set

Question 1

Consider an American put with possible early exercise. Define the state variables (spot tree, exercise payoff, continuation value, stopping time, and discount factor) and derive the optimal stopping recursion. State the measure, filtration, and tradability assumptions.

$$V_t = \max(g(S_t), e^{-r\Delta t} E^Q[V_{t+\Delta t} | \mathcal{F}_t])$$

Question 2

For an American put with possible early exercise, explain which probability measure is natural for pricing. Write the density-process logic needed if the starting model is under the physical measure.

Question 3

Identify the discounted tradable or density-adjusted process in an American put with possible early exercise. Prove the martingale condition and name one modelling choice that would break it.

Question 4

Using a three-step tree and an LSM simulation for a one-year put, construct a two-step numerical calculation. Show intermediate state values before discounting or aggregating.

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Question 5

For an American put with possible early exercise, compute and interpret early-exercise boundary and local delta around the boundary. Explain the hedge sign and the unit of each sensitivity.

Question 6

Suppose the desk observes basis functions for continuation value regression. Write the calibration objective, the weights you would use, and the parameter-identifiability checks.

Question 7

State the terminal and boundary conditions for an American put with possible early exercise. Explain how exercise region, continuation region, and terminal payoff prevents economically impossible prices.

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Question 8

Design a Monte Carlo or lattice estimator for an American put with possible early exercise. Specify Longstaff-Schwartz backward regression, the estimator, the discounting rule, and the standard error.

Question 9

A junior quant implements an American put with possible early exercise and gets a stable but wrong answer. Explain how perfect-foresight exercise decisions in Monte Carlo creates the error and design a diagnostic test.

Question 10

Write an NumPy algorithm for an American put with possible early exercise. Include inputs, outputs, diagnostics, and one benchmark case.

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Question 11

Starting from the payoff definition of an American put with possible early exercise, derive a replication, dynamic-programming, or conditional-expectation representation without skipping the economic argument.

Question 12

Create a stressed-market version of a three-step tree and an LSM simulation for a one-year put. Recompute the value or risk metric and explain which term dominates the change.

Question 13

Use a second-order Taylor approximation for an American put with possible early exercise. Separate linear exposure, convexity exposure, cross terms if relevant, and residual model risk.

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Question 14

Show how a change of numeraire or change of measure would simplify one expectation in an American put with possible early exercise. Identify what becomes a martingale after the change.

Question 15

Give a proof-style answer showing when the wealth process for an American put with possible early exercise is self-financing. Explain why predictable trading strategies matter.

Question 16

Market quotes for basis functions for continuation value regression are noisy. Propose a robust calibration workflow with initial guesses, constraints, residual plots, and out-of-sample checks.

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Question 17

For an American put with possible early exercise, compare the mathematical boundary condition with the trader's intuition. Give one limiting case where the answer should be obvious.

Question 18

Design a convergence study for Longstaff-Schwartz backward regression. Specify path counts, time-step refinement, confidence intervals, and the exact quantity plotted.

Question 19

Find one way that perfect-foresight exercise decisions in Monte Carlo could pass a superficial unit test. Write a stronger test that would fail for the biased implementation.

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Question 20

Explain an American put with possible early exercise and code the core routine. Give the derivation outline, pseudocode, and three numerical sanity checks.